

Duration: Half Hour

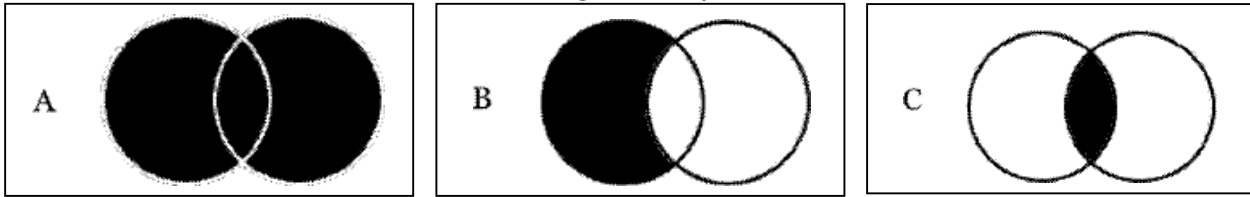
Onboarding Process

Ingen Dynamics - Programming Aptitude Assessment

- 1- What is the value of A if $A = 5 + (2 \times (9 + 5) - 4) / 2$
- 2- Your sister is 8 years old. You are three times as old as her. How old will you be when she is twice as old as her?
- 3- Which of the following is the odd one out: 3 ... 9 ... 12 ... 24 ... 8 ... 16
- 4- What is the value of the following expression: $(7 \bmod 5) + (4 \bmod 5) + (10 \bmod 5)$
- 5- A car rental service charges 200 E.P. a day and 1.5 E.P. a Kilometer to rent a car. Find the expression for total cost "C", in E.P., of renting a car for "D" days to travel "M" Kilometer.
- 6- One day Mrs. Aryl worked three-and-one-half hours in the morning, took a one-half hour lunch break, and worked four-and-one-half hours in the afternoon. If she began work at 8:30 a.m., what time did she finish?
- 7- Follow the procedures below:
 1. Put 5 in Box A.
 2. Put 4 in Box B.
 3. Add the number in Box A and the number in Box B and put the result in Box C.
 4. Add the number in Box A and the number in Box C and put the result in Box A.
 5. Write down the number from Box A, B, and C.
- 8- Solve the following:
 black sheep = dog
 white dog = tin
 bud black cow = dog
 white sheep = ?
- 9- Ali, Nour, and Body each have a favorite food: donuts, egg, and fish, in no particular order.
 List all of the ways in which you can choose two people and two foods. For example:
 Ali, Nour ; donuts, egg.
- 10- You're given 2 dice (die). List all of the possible combinations of the rolls that you might get, assuming order doesn't matter. For example: 1,2 and 2,1 Are the same combination.
- 11- You're given four colored boxes to be stacked one top of each other: red, green, blue, and yellow. Assume you must stack all four boxes, list each of the ways in which the boxes can be stacked.
- 12- The rate of growth of an expression is defined as how quickly the value of the expression increases when increasing values of n are used to evaluate the expression. Arrange these 7 mathematical expressions in order from slowest to fastest rates of growth:

$N^2 + 2n$	n^2	n^3	$2n$	n	5
	2^n				

13- The dark area of which of these diagrams represents "AND".



14- An ATM machine has 20 and 50 coins. Find The mathematical function for finding m , n as a function of x , such that $x=50m+20n$.

15- The company has information of their employers in three different lists.

You have

acquired all three lists, which all have a little bit different information depending on the

purpose of the list. The lists have the following information about the employees:

List 1: the number, the name, the occupation, and the department.

List 1 is ordered by the number of employees to be ascending order.

List 2: the name, the number, the address, the phone number, and the social security number

List 2 is organized to alphabetic order by the name.

List 3: the number, the social security number, salary, and some secret information

List 3 is ordered by the number to be ascending order.

Your job is to make a report on those employees whose salary is greater than 2000 E. P.

The report must display the name, the address, the department and the salary of the employees.

Describe how you would solve the problem. 16- Let i and j be integers.

List all values of i and j that make the expression to be always true a)

($i \geq 1$) and ($i \leq 5$)

The expression is true when i has values:

b) ($j \geq 3$) or ($j \leq 7$)

The expression is true when j has values:

17- Write a letter sequence, that is next on the series. Try to determine the general form of the series:

a) bce , bbcde , bbbcdde , _____ The

general form:

b) bcace , bcacacace , bcacacacacace , _____ The

general form:

c) _____, abccdd , abbccdd , abbbccdd , ... The

general form:

18- Your job is to sum up 50 numbers and at the end report the sum and the count of numbers that were positive numbers (>0).

Describe how you would solve the problem or write in a pseudo-code.

19- Your job is to deduce which of the given numbers is the smallest. How many comparisons you have to make if you have:

a) 3 numbers _____

b) 6 numbers _____

c) n numbers _____ ($n \geq 1$)
20- int a = 10; int b = 20;
what is the new value of a and b when:
a = b